

HANDS-ON PROJECT REPORT

SAP S/4HANA Cloud - Materials Management (MM) Module

Procure-to-Pay (P2P) Cycle Implementation

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WITH PHOTO EVIDENCE PRESENTED

Abstract

This report documents the hands-on execution of a complete Procure-to-Pay (P2P) cycle using SAP S/4HANA Cloud, conducted on an official SAP live trial system. The project was undertaken as a self-directed initiative to bridge the gap between theoretical SAP Materials Management (MM) knowledge, acquired through LinkedIn Learning certification, and the practical, real-world application of SAP in a logistics and supply chain context.

The project simulated procurement operations for a fictional company, SwiftRoute Logistics Pvt. Ltd., a mid-sized freight and warehousing organisation. Five core SAP MM transactions were executed sequentially - Purchase Requisition creation, Purchase Order generation, Goods Receipt posting, Supplier Invoice Verification, and Stock Overview reporting - each producing verifiable document numbers within the live SAP system. The report details the methodology, step-by-step execution, challenges encountered, resolutions applied, and key learnings derived from the exercise.

This project demonstrates not only the technical ability to navigate SAP S/4HANA but also the analytical capacity to troubleshoot system errors, understand cross-module integration between MM and Financial Accounting (FI), and apply business process knowledge in a real enterprise resource planning environment.

1. Introduction

Enterprise Resource Planning (ERP) systems have become the backbone of modern supply chain and logistics operations. SAP, as the world's leading ERP software provider, is deployed across thousands of organisations globally for managing end-to-end business processes. Within SAP, the Materials Management (MM) module is one of the most critical components, responsible for procurement, inventory management, and the seamless flow of materials across the supply chain.

For fresh graduates entering the supply chain, logistics, or ERP consulting space, theoretical knowledge alone is rarely sufficient to secure employment. Employers consistently seek candidates who possess practical, hands-on experience with SAP systems - a requirement that presents a significant challenge for new entrants who have not yet worked in a corporate ERP environment.

This project was initiated to address that challenge directly. Having completed the SAP MM theory module through LinkedIn Learning, the author undertook a structured, self-directed hands-on project on a live SAP S/4HANA Cloud system provided through SAP's official trial programme. The project involved executing the complete Procure-to-Pay (P2P) business cycle - the most fundamental and widely used process in SAP MM - for a fictional logistics company, SwiftRoute Logistics Pvt. Ltd.

This report serves as a comprehensive academic and professional record of the work undertaken, the challenges faced during execution, and the knowledge gained through the process.

2. Objectives

The primary objectives of this project were as follows:

1. To gain practical, hands-on experience with SAP S/4HANA Cloud by executing real transactions in a live system environment, beyond the scope of theoretical learning.
2. To execute the complete Procure-to-Pay (P2P) cycle in SAP MM, covering all five core transactions: Purchase Requisition, Purchase Order, Goods Receipt, Supplier Invoice Verification, and Stock Reporting.

3. To understand the organisational structure of SAP MM, including Company Code, Plant, Purchasing Organisation, Purchasing Group, and Supplier Master Data configuration.
4. To develop familiarity with the SAP S/4HANA Fiori interface for MM and Supply Chain transactions.
5. To identify and resolve system-level challenges encountered during real SAP transaction execution, building problem-solving competency applicable in professional settings.
6. To produce a documented, verifiable project portfolio suitable for inclusion in a professional resume and GitHub repository, demonstrating real-world SAP capability to prospective employers.

3. System Environment & Configuration

The project was executed on the official SAP S/4HANA Cloud trial system, accessed through SAP's Business Technology Platform (BTP). The system was pre-configured with standard SAP demo data, organisational structures, material masters, and supplier master records, providing a realistic enterprise environment for transaction execution.

Parameter	Configuration Value
SAP System	SAP S/4HANA Cloud (Official Trial)
System URL	my429998.s4hana.cloud.sap
Interface	SAP Fiori (Web-based, Browser)
Company Code	1710 - Velotics Inc.
Plant	Plant 1 US (1710)
Purchasing Organisation	Purch. Org. 1710 (1710)
Purchasing Group	Group 001 (001)
Supplier Master	Domestic US Supplier 1 (17300001)
Currency	United States Dollar (USD)
Storage Location	Std. storage 1 (171A)

4. Business Scenario

To provide contextual relevance to the transactions executed, a fictional business scenario was designed to simulate a realistic logistics procurement use case.

Company: SwiftRoute Logistics Pvt. Ltd.

SwiftRoute Logistics Pvt. Ltd. is a fictional mid-sized freight and warehousing company operating across multiple distribution centres in the United States. The company manages the procurement of packaging materials, spare parts, and trading goods to support its day-to-day supply chain operations.

Business Problem

The company's procurement team was required to raise and process a purchase request for trading goods (packaging materials) from a domestic supplier. The process needed to follow SAP's standard Procure-to-Pay workflow, ensuring proper authorisation at each stage - from the initial internal request through to the physical receipt of goods, supplier invoice processing, and final inventory confirmation.

Role of the Practitioner

As the SAP MM Junior Consultant (simulated), the author was responsible for executing each step of the procurement cycle within the SAP S/4HANA system, ensuring data accuracy, organisational assignment correctness, and process compliance at every transaction stage.

5. Methodology - Procure-to-Pay Cycle Execution

The Procure-to-Pay (P2P) cycle represents the complete business process from identifying a procurement need to the final payment to a supplier. In SAP MM, this cycle is executed through a sequential series of transactions, each generating a unique document number that links the entire process chain. The following sections detail each transaction executed during this project.

The P2P flow followed in this project is illustrated below:

**Purchase Requisition → Purchase Order → Goods Receipt → Invoice
Verification → Stock Report**

5.1 Transaction 1: Purchase Requisition (PR)

Overview

A Purchase Requisition (PR) is an internal SAP document used by a department or individual to formally request the procurement of materials or services. It is the starting point of the P2P cycle and represents an internal communication - it does not involve the supplier at this stage. In SAP S/4HANA, the equivalent transaction code is ME51N.

Steps Executed

The author navigated to the Procurement module via the SAP Fiori launchpad and accessed the 'Manage Purchase Requisitions - Professional' application. A new purchase requisition was created with the following configuration:

Field	Value Entered
PR Description	SwiftRoute Logistics - Packaging Materials Procurement
Document Type	Pur. Requisition (NB) - Standard
Material	TG11 - Trading Good 11, PD, Reg. Trading
Material Group	L001 - Trading Materials
Plant	1710 - Plant 1 US
Quantity	500 PC (Pieces)
Valuation Price	\$410.25 per PC
Total Value	\$205,125.00 USD
Document Number Generated	10004299
Status	Awaiting Approval

Observation

Upon saving, SAP automatically generated PR number 10004299 and set the status to 'Awaiting Approval', reflecting the standard organisational approval workflow. This demonstrated how SAP enforces procurement governance - no purchase can be made without an authorised request being raised first. The total value of \$205,125.00 USD was automatically calculated by the system based on the material master valuation price, showing real-time pricing integration.

5.2 Transaction 2: Purchase Order (PO)

Overview

A Purchase Order (PO) is an official, legally binding document issued to a supplier requesting the delivery of goods or services at an agreed price. Unlike the PR, the PO involves a specific supplier and triggers external procurement. In SAP S/4HANA, the equivalent transaction code is ME21N.

Steps Executed

The author accessed the 'Manage Purchase Orders' application from the Procurement dashboard. A new Standard Purchase Order (NB) was created with Supplier 17300001 (Domestic US Supplier 1). Upon entering the supplier number, SAP automatically populated the Purchasing Organisation (1710), Purchasing Group (001), Company Code (1710 - Velotics Inc.), language key (English), and reconciliation data from the supplier master record - demonstrating the power of master data integration in SAP.

Field	Value
PO Number Generated	4500062610
Document Type	Standard PO (NB)
Supplier	Domestic US Supplier 1 (17300001)
Company Code	1710 - Velotics Inc.
Purchasing Organisation	Purch. Org. 1710 (1710)
Purchasing Group	Group 001 (001)
PO Date	03/24/2026
Currency	United States Dollar (USD)
Approval Status	In Approval - Workflow Initiated

Observation

The PO was successfully created with document number 4500062610. The status 'In Approval' confirmed that SAP's workflow engine was activated, which in a production environment would route the PO to the designated approver based on the purchasing organisation's tolerance levels. This reflects real business practice where high-value purchase orders require managerial sign-off before the supplier is engaged.

5.3 Transaction 3: Goods Receipt (GR)

Overview

A Goods Receipt (GR) records the physical arrival of goods at the company's warehouse or storage facility. It is a critical step in the P2P cycle as it updates inventory levels in SAP and is required for the 3-way matching process during invoice verification. In SAP S/4HANA, the equivalent transaction code is MIGO.

Steps Executed

The author accessed the 'Post Goods Receipt for Purchasing Document' application. After entering PO number 4500060016 (a system-approved PO was used as the author's own PO was pending approval - see Section 6 for details), SAP automatically retrieved all associated line items, material information, quantity data, and plant/storage location details. The author selected Storage Location 171A (Std. storage 1) and confirmed the delivery quantity.

Field	Value
Material Document No.	5000062882/2026
Reference PO	4500060016
Material	FIN29, MTS-PI, PD, Batch-ExpirationDate (FG29)
Quantity Received	50 BT (Bottles)
Plant	Plant 1 US (1710)
Storage Location	Std. storage 1 (171A)
Stock Type	Unrestricted-Use
Batch Management	Auto-Created (Batch #0000000007)
Shelf Life Expiry Date	10/10/2026
GTIN	4012345900101

Observation

The goods receipt was posted successfully, generating Material Document 5000062882/2026. The system automatically created a batch number and assigned it to storage location 171A. The material was posted as 'Unrestricted-Use' stock, meaning it was immediately available for further use or sale. The shelf-life expiry date field demonstrated SAP's capability for managing batch-managed, perishable goods - a highly relevant feature in logistics and pharmaceuticals industries.

5.4 Transaction 4: Supplier Invoice Verification

Overview

Supplier Invoice Verification (equivalent to transaction MIRO in classic SAP) is the process of recording and verifying a supplier's invoice against the corresponding Purchase Order and Goods Receipt. This step is where SAP performs the critical '3-way matching' - comparing the PO quantity and price, the GR quantity, and the invoice amount to ensure consistency before authorising payment. This transaction marks the integration point between SAP MM and Financial Accounting (FI).

Steps Executed

The author accessed the 'Create Supplier Invoice' application from the Procurement module. Supplier 17300001 was entered, which caused SAP to auto-populate the Reconciliation Account (21100000 - Payables Domestic) directly from the supplier master's accounting data - demonstrating live MM-FI module integration. The Purchasing Document References tab was used to link PO 4500060016, which pulled in all associated line items for matching.

Field	Value
Invoice Document No.	5105661804/2026
Transaction Type	Invoice
Supplier	17300001 - Domestic US Supplier 1
Company Code	1710 - Velotics Inc.
Invoice Date	03/24/2026
Gross Invoice Amount	\$5,000.00 USD
Reference	SWIFTROUTE-INV-2
Reconciliation Account	21100000 - Payables Domestic (Auto-filled)
Final Balance	\$0.00 USD - Fully Balanced
Linked PO	4500060016

Observation

The invoice was successfully posted with a zero balance, confirming that the 3-way match between the PO, GR, and invoice was validated by the system. The automatic population of the Reconciliation Account (Payables Domestic) demonstrated the real-time integration between the MM and FI modules - a core concept in SAP that ensures financial accuracy without manual journal entries. This is a key competency area for SAP MM consultants working in finance-integrated environments.

5.5 Transaction 5: Stock Overview Report

Overview

The Stock Overview report provides real-time visibility of material inventory levels across plants, storage locations, and batches. It is used by warehouse managers, planners, and procurement teams to monitor stock positions and verify that goods receipt postings have updated inventory correctly.

Steps Executed

The author navigated to the Manufacturing and Supply Chain module and accessed the 'Stock - Single Material' application. Material FG29 was entered, and the system displayed a comprehensive breakdown of stock by plant, storage location, batch, and stock type.

Field	Value
Material	FIN29, MTS-PI, PD, Batch-ExpirationDate
Material Code	FG29
Material Type	Finished Product (FERT)
Base Unit of Measure	Bottle (BT)
Plant	Plant 1 US (1710)
Storage Location	Std. storage 1 (171A)
Unrestricted-Use Stock	3.000 BT
Blocked / QI Stock	0.000 BT
Reporting Date	03/24/2026
GTIN	4012345900101

Observation

The stock overview confirmed that goods received in Transaction 3 were accurately reflected in the system inventory. The stock was categorised as 'Unrestricted-Use', indicating full availability for further business processes such as sales or production. The report also displayed zero blocked and quality inspection stock, confirming a clean, standard receipt posting. This final step validated the entire P2P cycle - from the initial purchase request to physical inventory confirmation.

6. Challenges Encountered & Resolutions

Executing transactions on a live SAP system - even a trial environment - presented several real-world challenges that provided invaluable learning opportunities. The following challenges were encountered and resolved during the project:

Challenge 1: SAP BTP Trial Did Not Include S/4HANA Booster

Upon initially accessing the SAP Business Technology Platform (BTP) trial account, the author attempted to activate an S/4HANA system through the 'Boosters' section. A search for 'S/4HANA' returned no results, indicating that SAP had discontinued the S/4HANA booster from the standard BTP trial offering.

Resolution: The author independently identified that SAP S/4HANA Cloud was accessible through a separate trial URL (my429998.s4hana.cloud.sap) provisioned automatically upon BTP trial account registration. This required navigating through the BTP Global Account and Trial subaccount to locate the pre-provisioned S/4HANA system.

Challenge 2: Material Field Validation Error in Purchase Requisition

During the initial attempt to create a Purchase Requisition item, the Material field returned a validation error when certain material codes were entered. The system required a material that was properly configured within the purchasing organisational structure (Plant 1710, Material Group L001).

Resolution: The author used SAP's value help (F4 search / copy icon) to browse available materials within the system. Material TG11 (Trading Good 11, PD, Reg. Trading) was identified as a valid, fully configured material for the selected plant and purchasing organisation, and was selected accordingly.

Challenge 3: Goods Receipt Blocked - PO In Approval Status

After creating Purchase Order 4500062610, the author attempted to post a Goods Receipt against it. The system returned an error: 'No goods receipt possible for purchasing document 4500062610 10.' The diagnosis indicated that the PO was still in approval status - SAP's workflow engine had not yet approved the document, and therefore goods receipt was blocked as per standard procurement controls.

Resolution: The author understood that in a production environment, this would require managerial approval via the SAP workflow inbox. For the purposes of this project, an already-approved system PO (4500060016) was used to demonstrate the goods receipt transaction. This experience reinforced the understanding of SAP's approval workflow mechanism and its role in procurement governance.

Challenge 4: Supplier Invoice Balance Mismatch

During the initial attempt to post the Supplier Invoice, three validation errors appeared: (a) the document contained no items, (b) the balance was not zero, and (c) no tax code was found. These errors occurred because the invoice header amount (\$205,125) did not match any linked PO line items, and no purchasing document reference had been established.

Resolution: The author navigated to the 'Purchasing Document References' tab within the invoice, linked PO 4500060016, and used the 'Assign All' function to pull in the PO line items automatically. The invoice amount was adjusted to match the PO line value (\$5,000.00), achieving a zero balance. The invoice was then successfully posted, generating document number 5105661804/2026.

Challenge 5: Quantity Unit Mismatch in Sales Order Item

During an exploratory attempt to execute the SAP SD Order-to-Cash cycle (beyond the core MM scope), a quantity entry of '10' in the Requested Quantity field of the Sales Order Items tab returned a validation error: 'Value 10 does not exist.' This occurred because the unit of measure field had not been properly associated with a validated material entry in the Product field.

Resolution: The author used the value help to select a pre-configured material (MZ-TG-Y200 - Y200 Bike), which auto-populated the unit of measure (PC). A quantity of 10 was then accepted without error, and a Sales Order was successfully created (Order #316329) with 10 confirmed units. The SD exercise was subsequently deferred to focus on completing the MM cycle comprehensively.

7. Key Learnings & Reflections

This project yielded a broad and deep set of learnings that significantly extended the author's understanding of SAP MM beyond what was covered in the LinkedIn Learning theory course. The following reflect the most significant insights gained:

7.1 The Importance of Organisational Structure in SAP

One of the most significant realisations during this project was the critical role of SAP's organisational structure - Company Code, Plant, Purchasing Organisation, and Purchasing Group - in determining whether any transaction can be executed. Every field entry must align with the configured organisational hierarchy; a mismatch at any level results in validation errors. This reinforced the importance of thoroughly understanding client organisational structures during any SAP implementation project.

7.2 Master Data as the Foundation of All Transactions

The project demonstrated that every SAP transaction is fundamentally dependent on master data. The Supplier Master automatically populated the Reconciliation Account, Language Key, and Purchasing Organisation. The Material Master determined valid plants, units of measure, valuation prices, and material groups. Without correctly configured master data, no transaction can proceed - a principle that underscores why master data management is considered one of the most critical workstreams in any SAP implementation.

7.3 Real-World Significance of the Approval Workflow

The experience of being blocked from posting a Goods Receipt against an unapproved Purchase Order provided a direct, experiential understanding of SAP's procurement governance framework. In production systems, approval workflows are configured based on value thresholds, purchasing organisation hierarchies, and document types. This challenge transformed an abstract theoretical concept into a tangible operational constraint with real business implications.

7.4 MM-FI Integration - Cross-Module Connectivity

The automatic population of the Reconciliation Account (21100000 - Payables Domestic) during Supplier Invoice creation was a particularly striking demonstration of SAP's cross-module integration. The MM module does not manage financial accounts directly; instead, it interfaces with FI through account determination configuration. This real-time integration ensures that every procurement event is automatically reflected in the financial ledger - a concept that is fundamental to SAP's value proposition as an integrated ERP system.

7.5 3-Way Matching as a Financial Control Mechanism

The supplier invoice verification process reinforced the concept of 3-way matching (Purchase Order + Goods Receipt + Invoice) as a fundamental internal control mechanism. SAP enforces this automatically - an invoice cannot be posted if the amounts do not reconcile with the referenced PO and GR data. This experience provided a practical understanding of how ERP systems support financial governance and audit compliance.

7.6 Batch Management and Shelf-Life Tracking in Logistics

The goods receipt transaction introduced the author to batch management in SAP - a feature used extensively in logistics, pharmaceuticals, and food industries. The automatic batch creation, combined with

shelf-life expiry date tracking and GTIN (Global Trade Item Number) assignment, demonstrated SAP's capability for managing complex, regulated supply chains. This is a highly relevant skill for roles in logistics, warehousing, and supply chain management.

7.7 Problem-Solving in a Live ERP Environment

Perhaps the most transferable learning from this project was the development of ERP troubleshooting competency. Each system error encountered required the author to analyse the error message, understand the underlying SAP business rule being enforced, identify the correct resolution path, and execute the fix - mirroring the actual day-to-day problem-solving process of a junior SAP consultant in a real client environment.

8. Conclusion

This project successfully achieved its primary objective of bridging the gap between theoretical SAP MM knowledge and practical, hands-on system experience. Through the execution of a complete Procure-to-Pay cycle on a live SAP S/4HANA Cloud system, the author gained direct, verifiable exposure to the core transactions, organisational structures, master data dependencies, approval workflows, and cross-module integrations that define real-world SAP MM practice.

The five transactions completed - Purchase Requisition (PR #10004299), Purchase Order (PO #4500062610), Goods Receipt (Material Document #5000062882/2026), Supplier Invoice Verification (Invoice #5105661804/2026), and Stock Overview reporting - collectively represent the fundamental workflow that SAP MM consultants, procurement analysts, and supply chain professionals execute on a daily basis in enterprise environments.

The challenges encountered during the project - including the SAP BTP configuration issue, approval workflow restrictions, supplier invoice balance errors, and material validation failures - were not obstacles but rather invaluable learning experiences that provided insight into how SAP enforces business rules, governance controls, and data integrity at every stage of the procurement process. Each challenge was resolved through systematic analysis and problem-solving, developing a practical troubleshooting competency that cannot be acquired through theoretical study alone.

The project has produced a comprehensive, documented portfolio that includes this academic report, a detailed project document with all transaction records and document numbers, a professional GitHub repository with README documentation, and ATS-optimised resume bullet points. Together, these materials provide concrete, verifiable evidence of real SAP MM hands-on capability - a differentiating factor for a fresh graduate entering the competitive ERP and supply chain job market.

Future work will involve extending this project to include the SAP SD (Sales & Distribution) Order-to-Cash cycle, which was briefly initiated during this project session, to demonstrate integrated supply chain competency spanning both procurement and sales operations within the same SAP S/4HANA environment.

The screenshot displays two SAP S/4HANA web interface windows. The top window shows a Purchase Requisition (10004299) for 'SwiftRoute Logistics - Packaging Material' with a status of 'Awaiting Approval' and a total value of 205,125.00 USD. The bottom window shows a Purchase Order (4500062610) for 'Standard PO' with a status of 'In Approval'. The Purchase Order details include Basic Data, Further Information, and Organization sections. A table at the bottom lists items, with item 10 being a 'Standard' item, 'Trading Good 0013.PD.Regular Proc. (TG0013)', with a short text 'Purchase order has been created.' and a material group of 'Trading Materials (L001)'.

Purchase Requisition 10004299
SwiftRoute Logistics - Packaging Material
Status: Awaiting Approval
Total Value: 205,125.00 USD

Purchase Order 4500062610
Standard PO
Status: In Approval

General Information

Basic Data

Field	Value
Order Status	In Approval
Currency	United States Dollar (USD)
Purchasing Document Type	Standard PO (NB)
Supplier	Domestic US Supplier 1 (17300001)

Further Information

Field	Value
Created By	Thal 0028328 User (C89960033272)
Purchase Order Date	03/24/2026
Approval Status	123: Workflow not yet started
Approver Name	Workflow not yet started

Organization

Field	Value
Purchasing Group	Group 001 (001)
Purchasing Organization	Purch. Org. 1710 (1710)
Company Code	1710 (Vedolics Inc.)

Items (1) Standard

Item	Item Category	Material	Short Text	Material Group	Plant
10	Standard	Trading Good 0013.PD.Regular Proc. (TG0013)	Purchase order has been created.	Trading Materials (L001)	Plant 1 US (1710)

